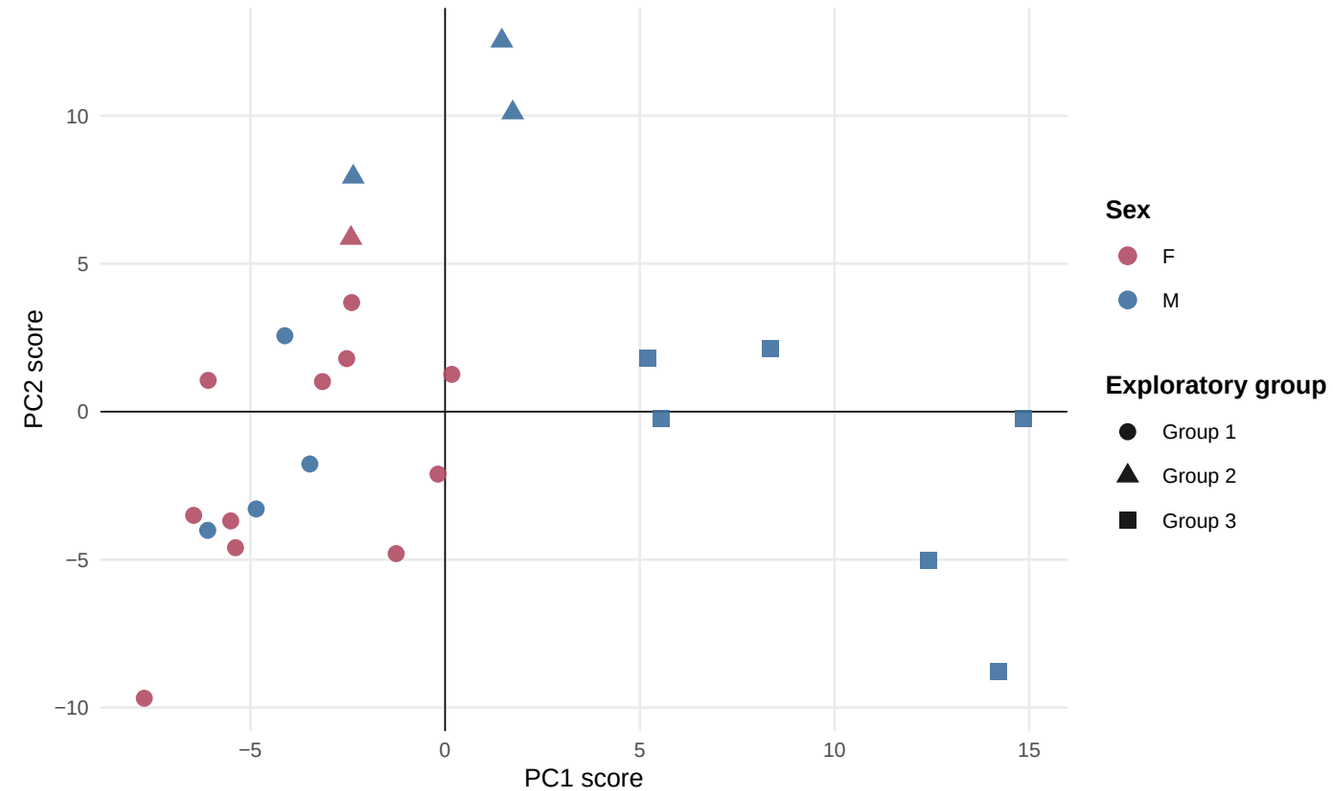
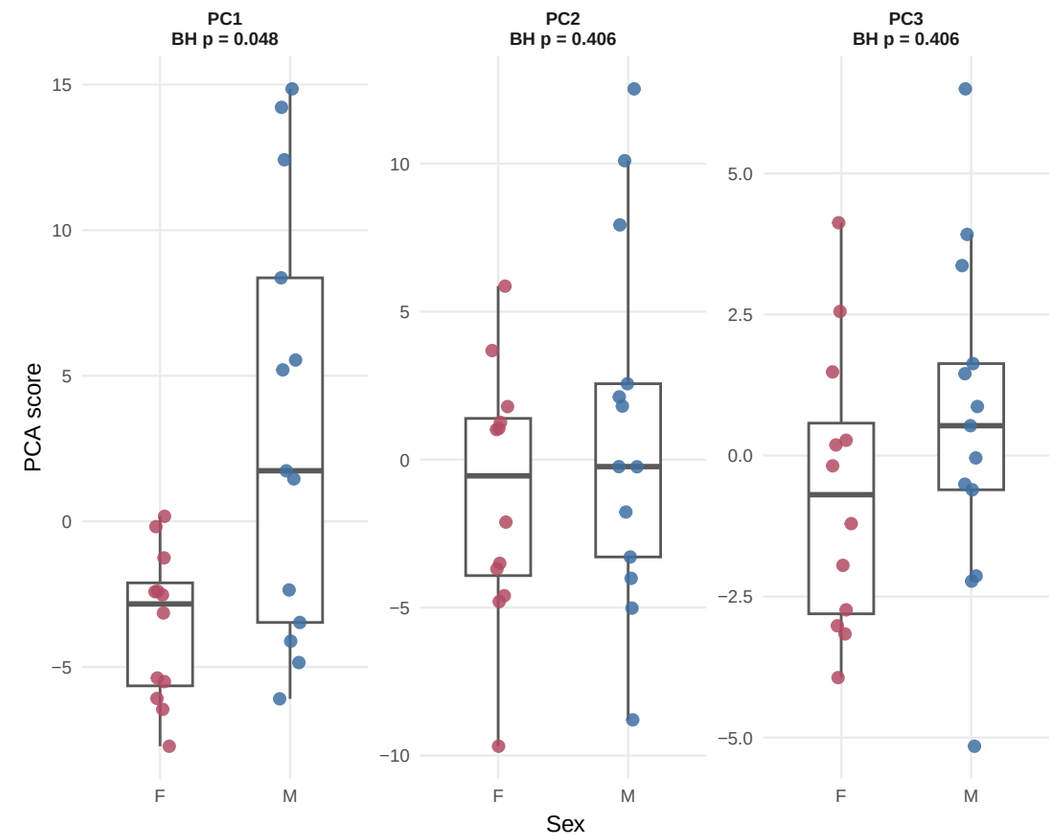
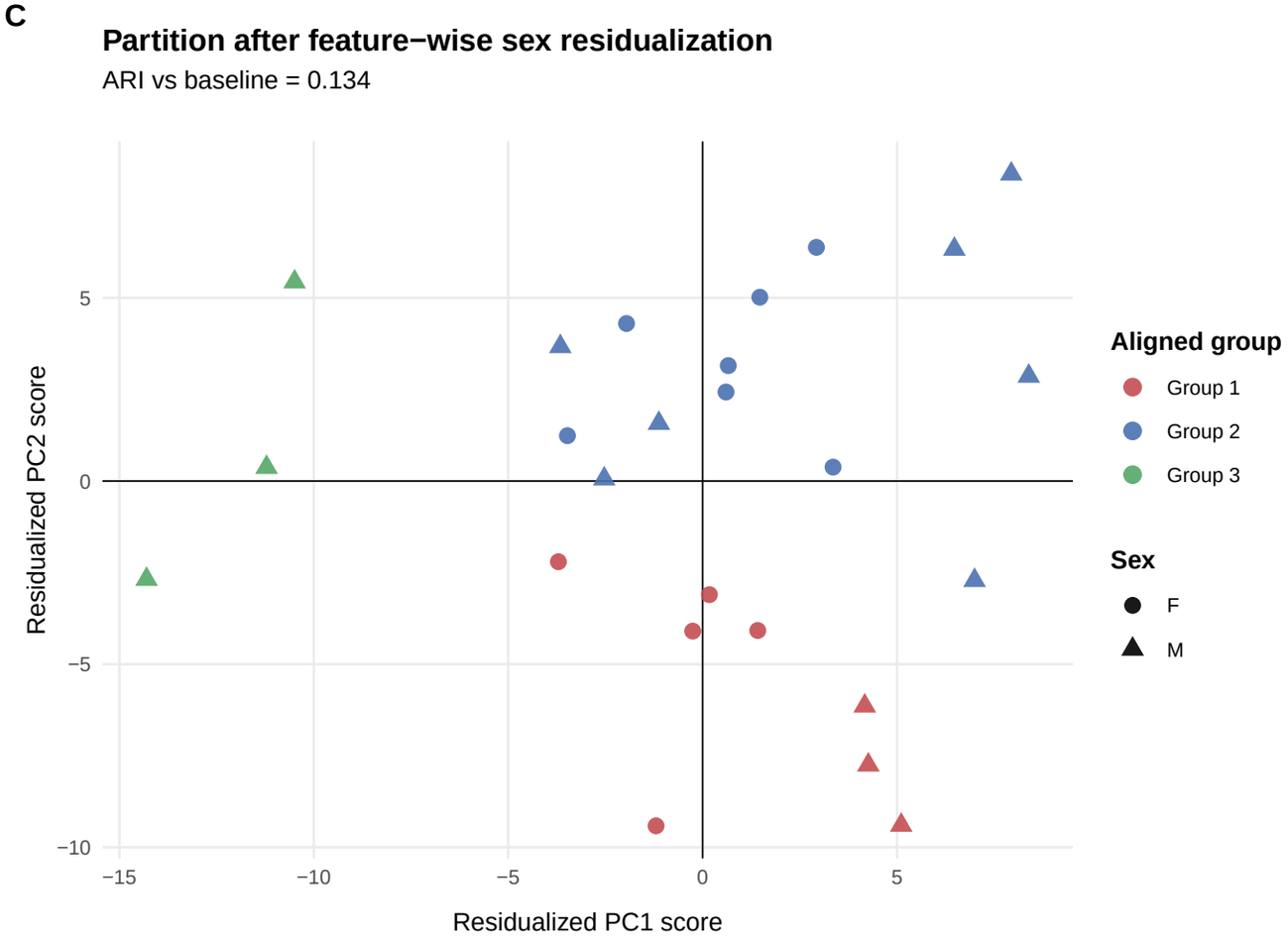
A


Baseline PCA scores by sex
PC2 score
PC1 score
10
5
0
−5
−10
−5
0
5
10
15
Sex
F
M
Exploratory group
Group 1
Group 2
Group 3

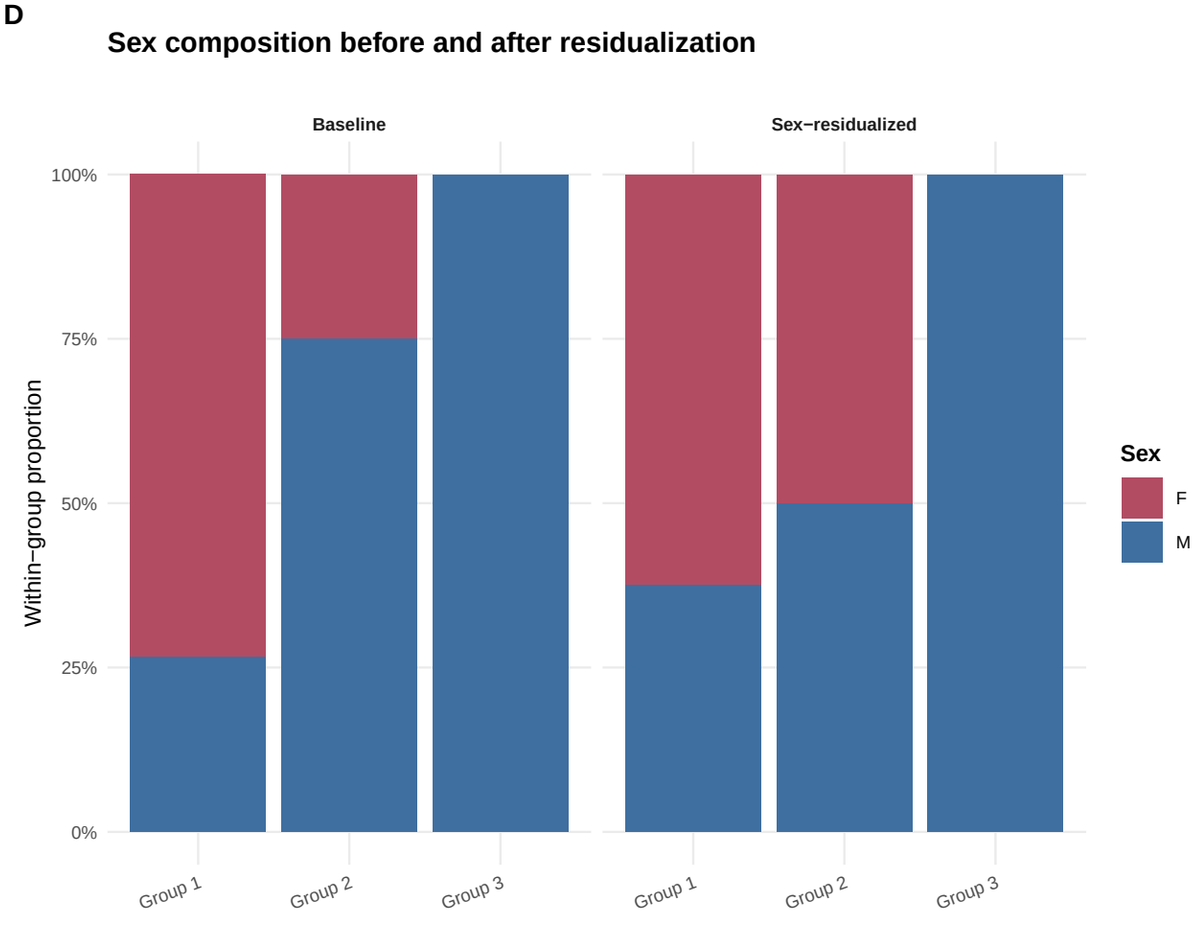
B


PC1−PC3 score distributions
PC1
BH p = 0.048
PC2
BH p = 0.406
PC3
BH p = 0.406
PCA score
Sex
F
M

C


Partition after feature−wise sex residualization
ARI vs baseline = 0.134
Residualized PC2 score
Residualized PC1 score
5
0
−5
−10
−15
−10
−5
0
5
Aligned group
Group 1
Group 2
Group 3
Sex
F
M

D


Sex composition before and after residualization
Baseline
Sex−residualized
Within−group proportion
100%
75%
50%
25%
0%
Group 1
Group 2
Group 3
Sex
F
M